

Aditya Dasika

Multi-Agent Systems on LangGraph · Graph RAG & FAISS Vector Search · LLM Evaluation, OpenTelemetry & Tool-Calling Guardrails

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EXPERIENCE

AI/LLM Engineer II

April 2025 – Present

Phenom People

Hyderabad, Telangana

- Owned the architecture of **chatbotCXAgent**, a production multi-agent system on **LangGraph** (12 orchestrated nodes) spanning intent recognition, planning, dialogue-state management and tool routing — with **Graph RAG** retrieval over enterprise knowledge and document repositories via FAISS vector search and a Neo4j graph for multi-hop, policy-aware grounding.
- Engineered tool-calling agents that execute governed actions — database queries and structured extraction from unstructured documents — under **XGrammar constrained decoding** and schema-validated JSON guardrails, raising tool-routing accuracy to **91.9%** and eliminating malformed tool calls in live traffic.
- Built an **LLM evaluation framework** (pass@k, LLM-as-judge rubric) gating every release across all 12 nodes, instrumented with **LiteLLM + Langfuse** for per-node cost, latency and full-trace auditability — continuously monitoring agent behaviour and curbing hallucinations.
- Ran the self-hosted LLM inference service as a deployable platform — Helm-packaged releases, custom chat templates per model, reasoning-model flags, and **OpenTelemetry** instrumentation feeding a metrics dashboard so per-model latency and throughput were visible in production.
- Building the detection and decisioning layer of an enterprise **hiring-integrity / fraud-risk platform** — an **LLM orchestration engine** that investigates flagged activity, correlates evidence across signals and produces reviewer-facing assessments with supporting rationale — backed by a **policy-driven rules engine**, per-tenant risk configuration and audited review workflows across the signal lifecycle.
- Designed the configuration and policy plane for that platform: a global fraud-signal registry with feature groups, tenant-scoped overrides and applicable-context gating, exposed through a **resolve API** consumed by the reviewer console and the live detection pipeline — including a role-gated signal-lifecycle state machine and versioned definitions that keep downstream consumers stable across config changes.
- Hardened the fine-tuned Qwen deployment for production readiness — a custom Jinja2 chat template, required tool-choice and temperature pinned at 0.1 for deterministic tool calls — then held it to a golden baseline of 1.0s median and **1.4s** P95 latency with zero failures across **5,510** requests, taking the service's test coverage from **7% to 78%**.
- Fine-tuned and served Qwen models via **QLoRA** (Unsloth, NF4) on **vLLM** / AWS g5.xlarge (A10G), sustaining 10.98 RPS at 98.6% success under load at roughly half the inference cost of hosted GPT-4.1.
- Built conversation analytics APIs covering both live-agent and WhatsApp channels, raised the service's automated testing coverage from **63% to over 90%** while clearing static-analysis code smells, and migrated the service to **FastAPI**.

Full Stack Engineer

May 2022 – February 2025

Blueleaves Farms

Hyderabad, Telangana

- Owned end-to-end delivery of production systems (POS, CRM, inventory, procurement, e-commerce) on Node.js, Firebase Cloud Functions and REST APIs — with role-based access control and CI/CD — integrating multiple internal systems and cutting manual operational errors by roughly 90%.

PROJECTS

IndexNotes — **AI Knowledge Graph for Scientific Papers** | *Neo4j, Ollama, PyMuPDF, Python, Docker*

- Built an automated Graph RAG pipeline ingesting research papers (PDF → local-LLM extraction → structured JSON → Neo4j), creating 40+ concepts and 17+ typed relationships, with quantized local models via Ollama for entity/relationship extraction, on a Neo4j schema enabling multi-hop graph traversal and cross-document retrieval.

TECHNICAL SKILLS

GenAI & Agentic: Agentic / Multi-Agent Systems, LangGraph, LangChain, Tool/Function Calling, Prompt Engineering, RAG & Graph RAG, Model Routing, Context-Window Management, LLM Fine-Tuning (QLoRA, SFT)

Evaluation, Observability & Governance: LLM Evaluation (LLM-as-judge, pass@k), OpenTelemetry, Langfuse, LiteLLM, Model Monitoring, Constrained Decoding (XGrammar), Schema Validation, Hallucination Mitigation, Role-Based Access Control

Vector & Graph Retrieval: Embeddings / Vector Search (FAISS), Neo4j, MongoDB, PostgreSQL, ClickHouse, Redis

Backend, Platform & Delivery: Python, TypeScript, FastAPI, Node.js, REST APIs, Microservices, Docker, Helm, Temporal, Kafka, AWS (EC2/S3/SageMaker), GitHub Actions, CI/CD, Linux

ML & Serving: PyTorch, vLLM, Transformers (BERT/ModernBERT), HuggingFace PEFT, Unsloth, Ollama, Production MLOps

EDUCATION

Andhra Loyola Institute of Engineering and Technology

Bachelor of Technology in Mechanical Engineering

Vijayawada, Andhra Pradesh

August 2018 – May 2022